1 Inputs

Parameter	Value	Unit	Source
Base parameter set	Chen2020 (NMC811/graphite)	-	entry 0 meta.base (anchor table; task names no electrode system)
Negative particle radius	1.5e-06	m	params_v11.json (design)
Positive particle radius	3.5e-06	m	params_v11.json (design)
Electrolyte conductivity	2.5	S/m	params_v11.json (design)
Cation transference number	0.55	-	params_v11.json (design)
Electrolyte diffusivity	6e-10	m2/s	params_v11.json (design)
Negative electrode thickness	0.000115	m	params_v11.json (design)
Negative electrode porosity	0.36	-	params_v11.json (design)
Total heat transfer coefficient h	50	W/(m2 K)	params_v11.json (design)
Nominal cell capacity	6.0313	Ah	params_v11.json (= measured 1C, mechanical rule propose_r2b)
Positive electrode thickness	7.56e-05	m	Chen2020 parameter set
Positive electrode porosity	0.335	-	Chen2020 parameter set
Separator thickness / porosity	12e-6 / 0.47	m / -	Chen2020 parameter set
Electrode height x width	0.065 x 1.58	m	Chen2020 parameter set
Electrode area	0.1027	m2	height x width = 0.1027 (matches calc-energy area_m2)
Voltage window	2.5 - 4.2	V	Chen2020 parameter set (cut-offs)
Max conc positive / negative	63104 / 33133	mol/m3	Chen2020 parameter set
Positive/negative electrode density	3262 / 1657	kg/m3	Chen2020 parameter set
Pos/neg CC thickness + density	16um Al 2700 / 12um Cu 8960	-	Chen2020 parameter set
SEI kinetic rate constant	1e-12	m/s	Chen2020 parameter set

2 Capacity-Energy

Quantity	Value	Unit	Formula / source
Measured 1C capacity (DFN)	6.032209	Ah	cell/r5_v11_1c_dfn.json:capacity_ah
Discharge energy	21.796964	Wh	cell/r5_v11_energy.json:energy_wh (integral V x I over 1C discharge)
Midpoint voltage	3.723722	V	cell/r5_v11_energy.json:midpoint_voltage_v
Average voltage	3.6135	V	energy / capacity = 21.7970 / 6.0322 (mechanical)
DC resistance	0.00288198	ohm	cell/r5_v11_energy.json:dcr_ohm
Power density	32559.66	W/kg	cell/r5_v11_energy.json:power_density_w_kg
Areal capacity (cell)	58.7356	Ah/m2	6.032209 Ah / 0.1027 m2

3 Energy Density

Quantity	Value	Unit	Formula / source
Positive coating mass	16.842	g	layer_kg_m2 0.163994 x 0.1027 (r5_v11_energy.json)
Negative coating mass	12.525	g	layer_kg_m2 0.121955 x 0.1027
Al current collector	4.437	g	layer_kg_m2 0.0432 x 0.1027
Cu current collector	11.042	g	layer_kg_m2 0.10752 x 0.1027
Separator	0.259	g	layer_kg_m2 0.00252492 x 0.1027
Total mass (contract, electrolyte excluded)	45.105	g	cell/r5_v11_energy.json:mass_kg = 0.0451052 kg
Gravimetric energy density	483.247	Wh/kg	21.796964 Wh / 0.0451052 kg -> threshold 327.18 PASS
Cell volume	2.368262e-05	m3	cell/r5_v11_energy.json:volume_m3
Volumetric energy density	920.378	Wh/L	21.796964 Wh / 23.68262 cm3
Stack thickness	0.0002306	m	75.6 + 12 + 115 + 16 + 12 um (cell/r5_v11_energy.json:thickness_m)
Electrolyte mass (estimate, excluded)	8.918	g	pore volume 7.432 cm3 x 1.2 g/cm3 (literature)
Reference ED incl. electrolyte (not adjudicated)	403.47	Wh/kg	21.796964 Wh / 0.0540234 kg (reference only)

4 N-P and Mass

Quantity	Value	Unit	Formula / source
Negative theoretical areal capacity	65.357	Ah/m2	115e-6 x (1-0.36) x 33133 mol/m3 x 96485/3600
Positive theoretical areal capacity (full range)	85.027	Ah/m2	75.6e-6 x (1-0.335) x 63104 mol/m3 x 96485/3600
Cell measured areal capacity	58.7356	Ah/m2	6.032209 Ah / 0.1027 m2 (= positive usable capacity)
Practical N/P ratio	1.1127	-	65.357 / 58.7356 (negative theoretical / positive usable)
Theoretical full-range N/P (reference)	0.7687	-	65.357 / 85.027 (positive full Li range never used below 4.2 V)
Positive utilization in 2.5-4.2 V window	0.6908	-	58.7356 / 85.027

5 Process

Parameter	Value	Unit	Formula / source
Positive areal density	164	g/m2	75.6e-6 x (1-0.335) x 3262 (design-spec process formula)
Negative areal density	122	g/m2	115e-6 x (1-0.36) x 1657
Positive compaction density	2.17	g/cm3	3262 x 0.665 / 1000
Negative compaction density	1.06	g/cm3	1657 x 0.64 / 1000
Electrolyte fill amount	8.92	g	7.432 cm3 x 1.2 g/cm3 x 1.0 fill factor (literature density; fill factor assumption)
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	-	design recommended value; production value requires tuning